VAGUS NERVE

VAGUS NERVE

- It is a mixed nerve, but mostly it is motor.
- Vagus nerve is the 10th **cranial nerve**.
- Its innervation is not limited to the structures in the head, but also extends into the neck, thorax, and abdomen.
- It is the longest and most widely distributed cranial nerve.
- It is a wandering nerve.
- It spreads the fibres of a cranial part of the **accessory nerve** and carries majority of the efferent fibres of the cranial part of the parasympathetic outflow.

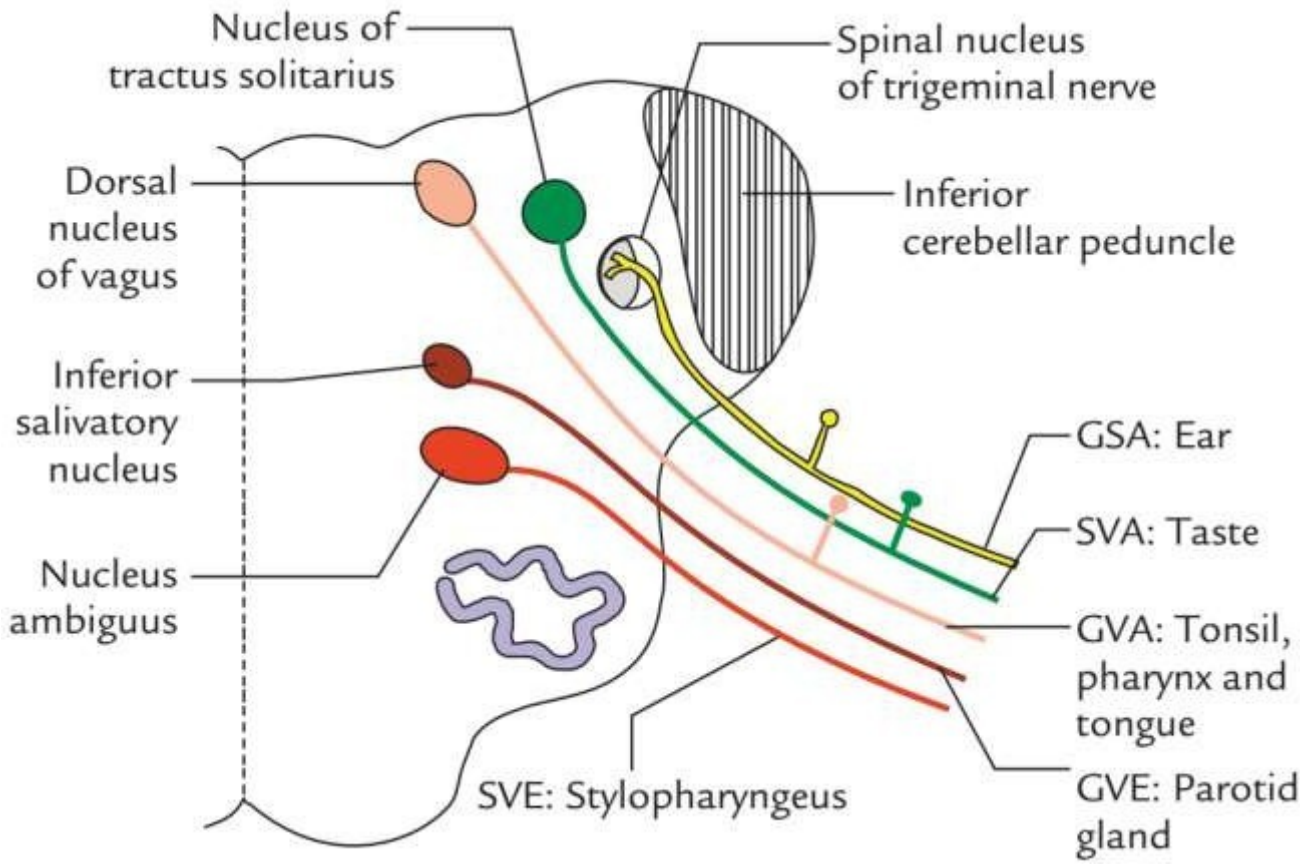
VAGUS NERVE

- The vagus nerve is associated with the derivatives of the fourth pharyngeal arch.
- **Sensory:** Innervates the skin of the external acoustic meatus and the internal surfaces of the laryngopharynx and larynx. Provides visceral sensation to the heart and abdominal viscera.
- **Special Sensory:** Provides taste sensation to the epiglottis and root of the tongue.
- **Motor:** Provides motor innervation to the majority of the muscles of the pharynx, soft palate and larynx.
- **Parasympathetic:** Innervates the smooth muscle of the trachea, bronchi and gastro-intestinal tract and regulates heart rhythm.

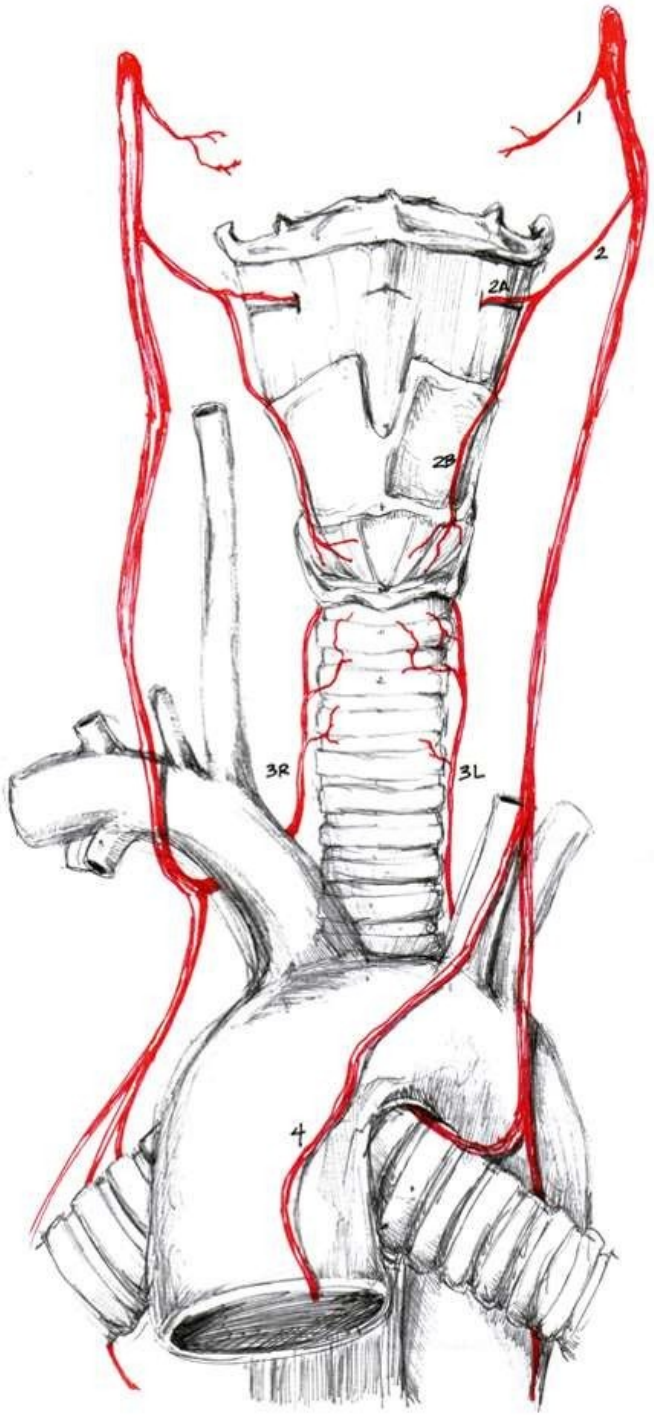
Types of fibers in vagus

- **A) Branchial motor**
- Supplying muscles of the pharynx and larynx.
- **B) Visceral sensory**
- This component of the vagus nerve is responsible for transmitting information from a wide variety of anatomical sites including the heart and lungs, pharynx and larynx, and upper part of the gastrointestinal tract.
- **C) Visceral motor**
- The visceral motor component carries parasympathetic fibers from the smooth muscle of the upper respiratory tract, heart, and gastrointestinal tract.
- **D) Special sensory**
- The special sensation conveyed by the vagus nerve is for taste from the palate and epiglottis.
- **E) General sensory**
- The general sensory component of the vagus nerve is concerned with information from parts of the ear and the dura within the posterior cranial fossa.

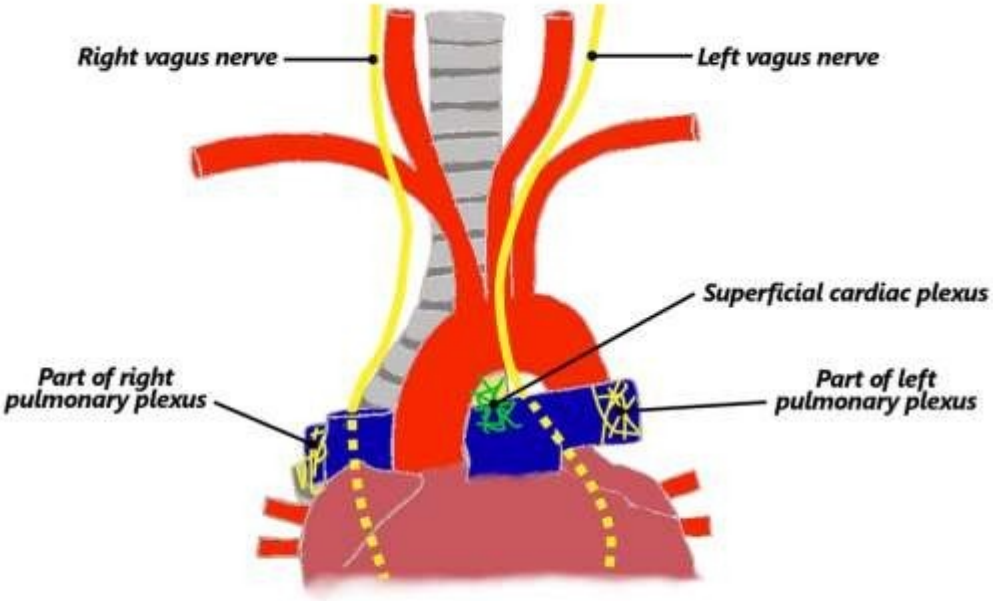
Vagus Nerve: Functional Parts and Nuclei

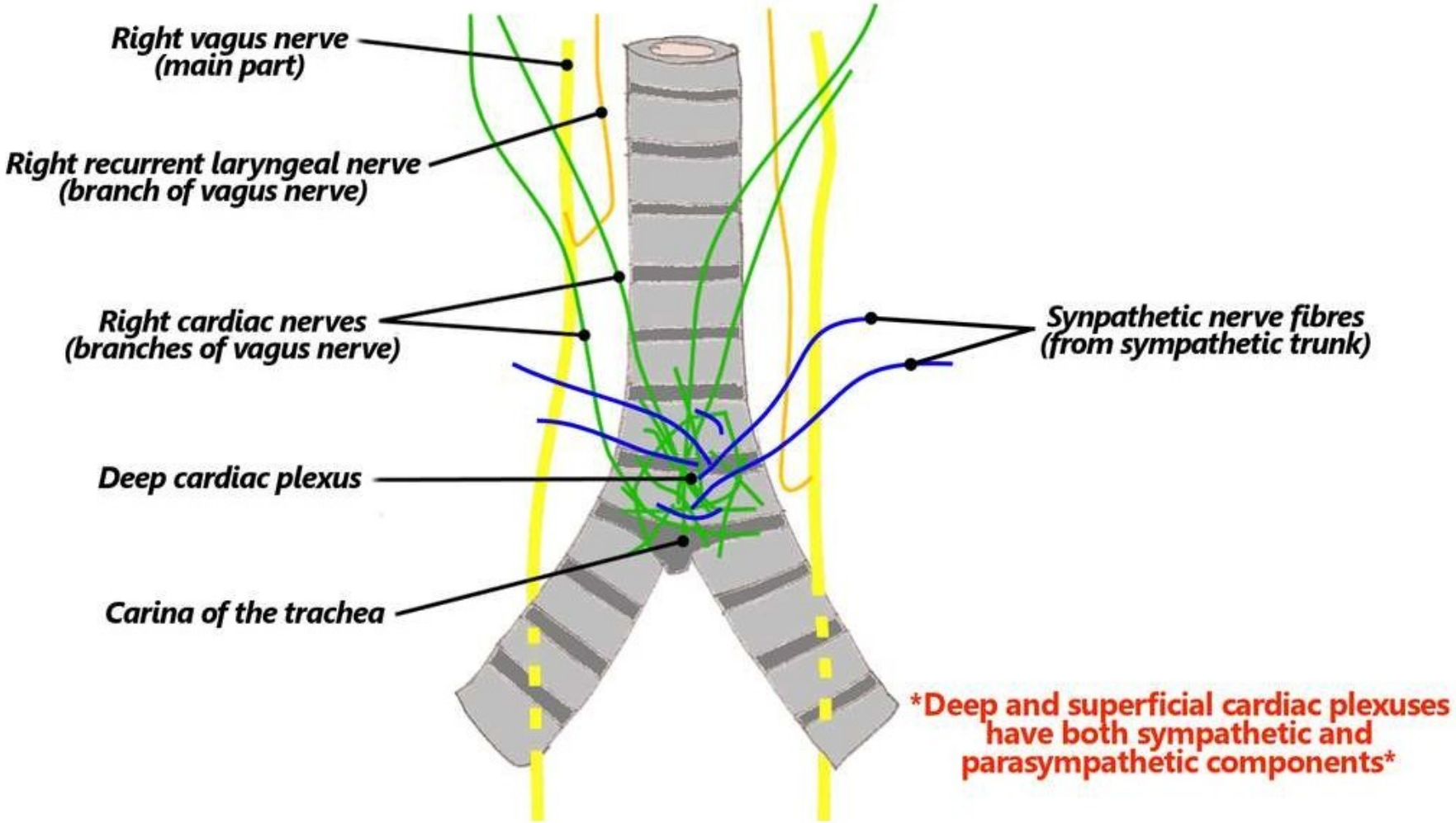


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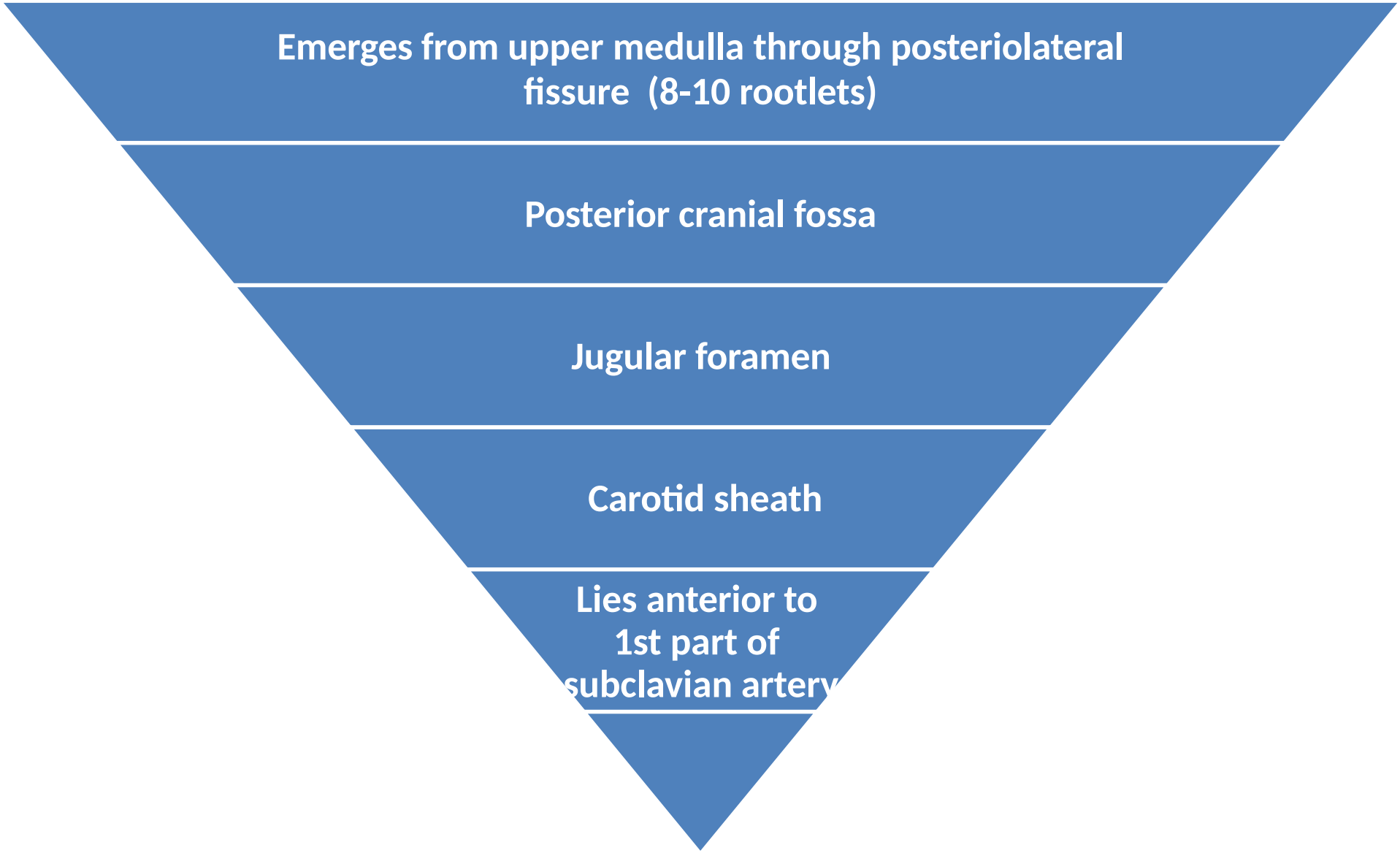


1 Pharyngeal branches
2 Superior Laryneal n.
2A Internal branch
2B External branch
3R Right Recurrent Laryngeal n.
3L Left Recurrent Laryngeal n.
4 Cervical Cardiac branches



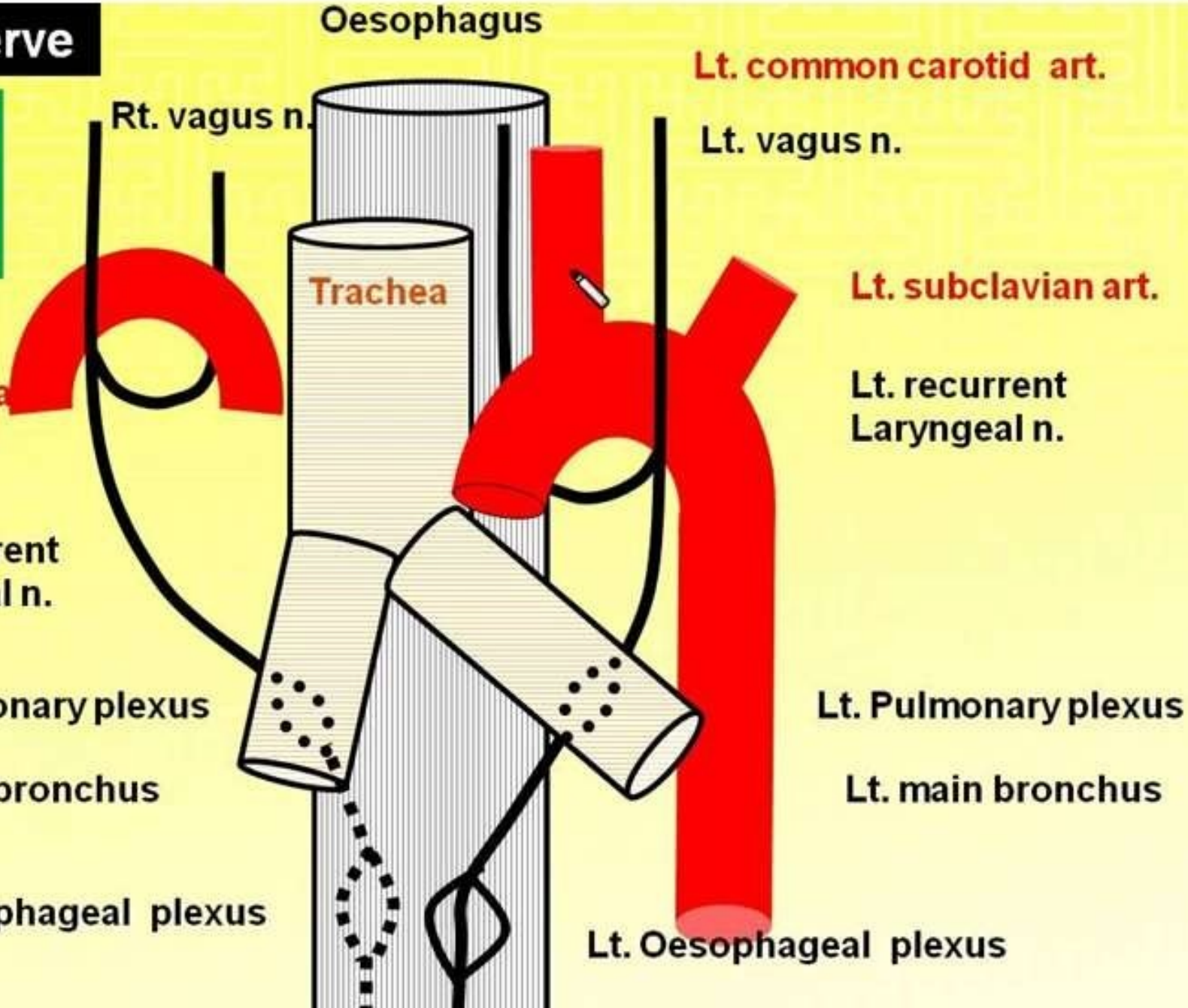

Right vagus nerve
(main part)
Right recurrent laryngeal nerve
(branch of vagus nerve)
Right cardiac nerves
(branches of vagus nerve)
Deep cardiac plexus
Carina of the trachea
Synpathetic nerve fibres
(from sympathetic trunk)
Deep and superficial cardiac plexuses have both sympathetic and parasympathetic components

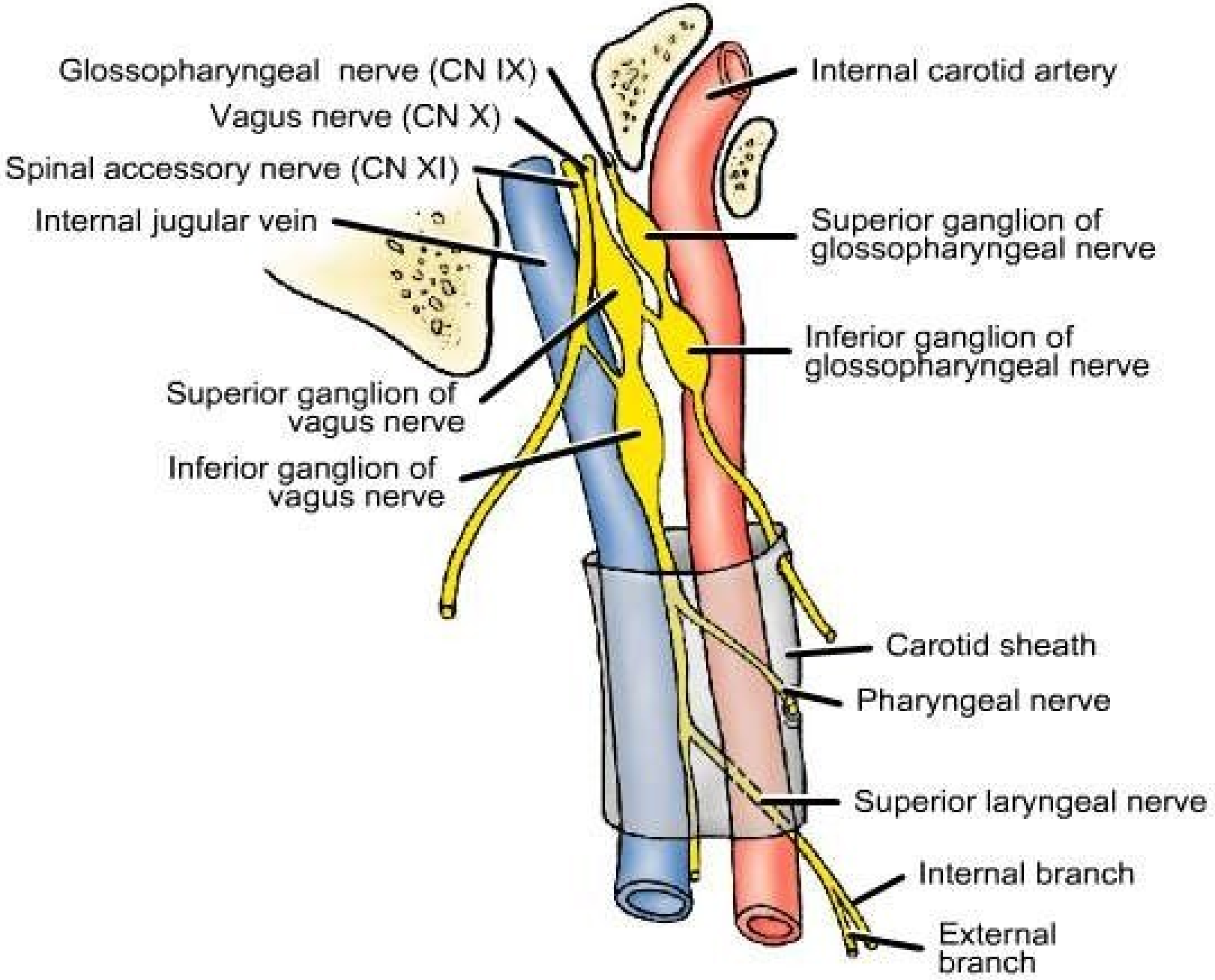
The Vagus Nerve (CN X)



Vagus nerve

Origin
Course
End
Branches



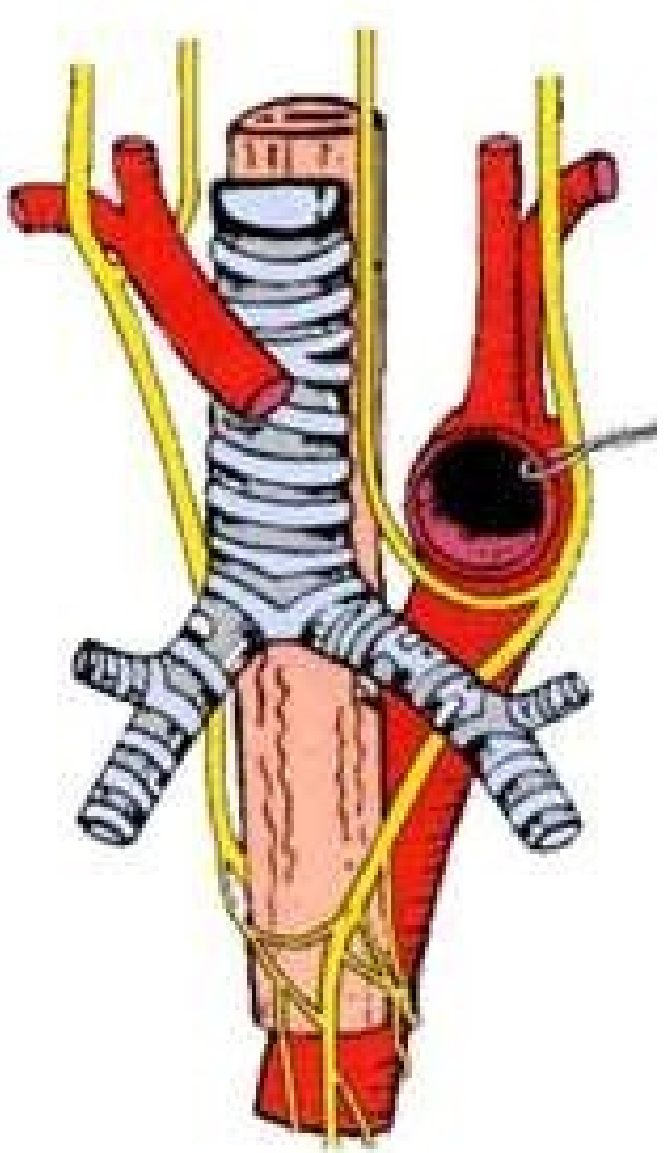

Glossopharyngeal nerve (CN IX)
Vagus nerve (CN X)
Spinal accessory nerve (CN XI)
Internal jugular vein
Superior ganglion of vagus nerve
Inferior ganglion of vagus nerve
Internal carotid artery
Superior ganglion of glossopharyngeal nerve
Inferior ganglion of glossopharyngeal nerve
Carotid sheath
Pharyngeal nerve
Superior laryngeal nerve
Internal branch
External branch

Right vagus

- In thorax first posterolateral to brachiocephalic artery
- Between trachea (laterally) azygos vein (medially)
- Behind root of lung(pulmonary plexus)
- Posterior to esophagus (esophageal plexus)
- Esophageal opening of diaphragm at T10
- Posterior surface of stomach to form posterior vagal trunk
- Branches to celiac & superior mesenteric plexuses

Right vagus nerve

- Enter thoracic inlet on right side of trachea
- Travels downward posterior to right brachiocephalic vein and superior vena cava
- Passes posterior to right lung root
- Forms posterior esophageal plexus
- Forms posterior vagal trunk at esophageal hiatus where it leaves thorax and passes into abdominal cavity, then divides into posterior gastric and celiac branches



Left vagus nerve

- X • Descends in thorax between left common carotid artery & left subclavian artery
- X • Crosses left side of aortic arch (phrenic nerve crosses left vagus)
- X • Behind root of left lung (pulmonary plexus)
- X • Anterior to esophagus (esophageal plexus)
- X • Esophageal opening of diaphragm at T10
- X • Anterior surface of stomach to form anterior vagal trunk

Left vagus nerve

- Enter thoracic inlet between left common carotid and left subclavian arteries, posterior to left brachiocephalic vein
- Crosses aortic arch where left recurrent laryngeal nerve branches off
- Passes posterior to left lung root
- Forms anterior esophageal plexus
- Forms anterior vagal trunk at esophageal hiatus where it leaves thorax and passes into abdominal cavity , then divides into anterior gastric and hepatic branches



VAGUS NERVE

- **In the Head**
- The vagus nerve originates from the lateral aspect of the medulla between the olive and inferior cerebellar peduncle by about 10 rootlets below and in line of the rootlets of the **glossopharyngeal nerve**.
- It exits the cranium via the **jugular foramen,** through the middle part of the jugular foramen with the glossopharyngeal and accessory nerves (CN IX and XI respectively).
- Within the cranium, the **auricular branch** arises.
  - This supplies sensation to the posterior part of the external auditory canal and external ear.
- **Meningeal branch:**
  - It appears from the superior ganglion, takes a recurrent course and enters the cranial cavity via the jugular foramen to supply the **dura mater** of the **posterior cranial fossa**.

VAGUS NERVE

- **In the neck**, the vagus nerve passes into the carotid sheath, travelling inferiorly with the internal jugular vein and common carotid artery.
- At the base of the neck, the right and left nerves have differing pathways:
  - The **right vagus nerve** passes anterior to the subclavian artery and posterior to the sternoclavicular joint, entering the thorax.
  - The **left vagus nerve** passes inferiorly between the left common carotid and left subclavian arteries, posterior to the sternoclavicular joint, entering the thorax.

VAGUS NERVE

- Several branches arise in the neck:
- **Pharyngeal branches** – Provides motor innervation to the majority of the muscles of the pharynx and soft palate.
- **Superior laryngeal nerve** – Splits into internal and external branches. The external laryngeal nerve innervates the cricothyroid muscle of the larynx. The internal laryngeal provides sensory innervation to the laryngopharynx and superior part of the larynx.
- **Recurrent laryngeal nerve** (right side only) – Hooks underneath the right subclavian artery, then ascends towards to the larynx. It innervates the majority of the intrinsic muscles of the larynx.

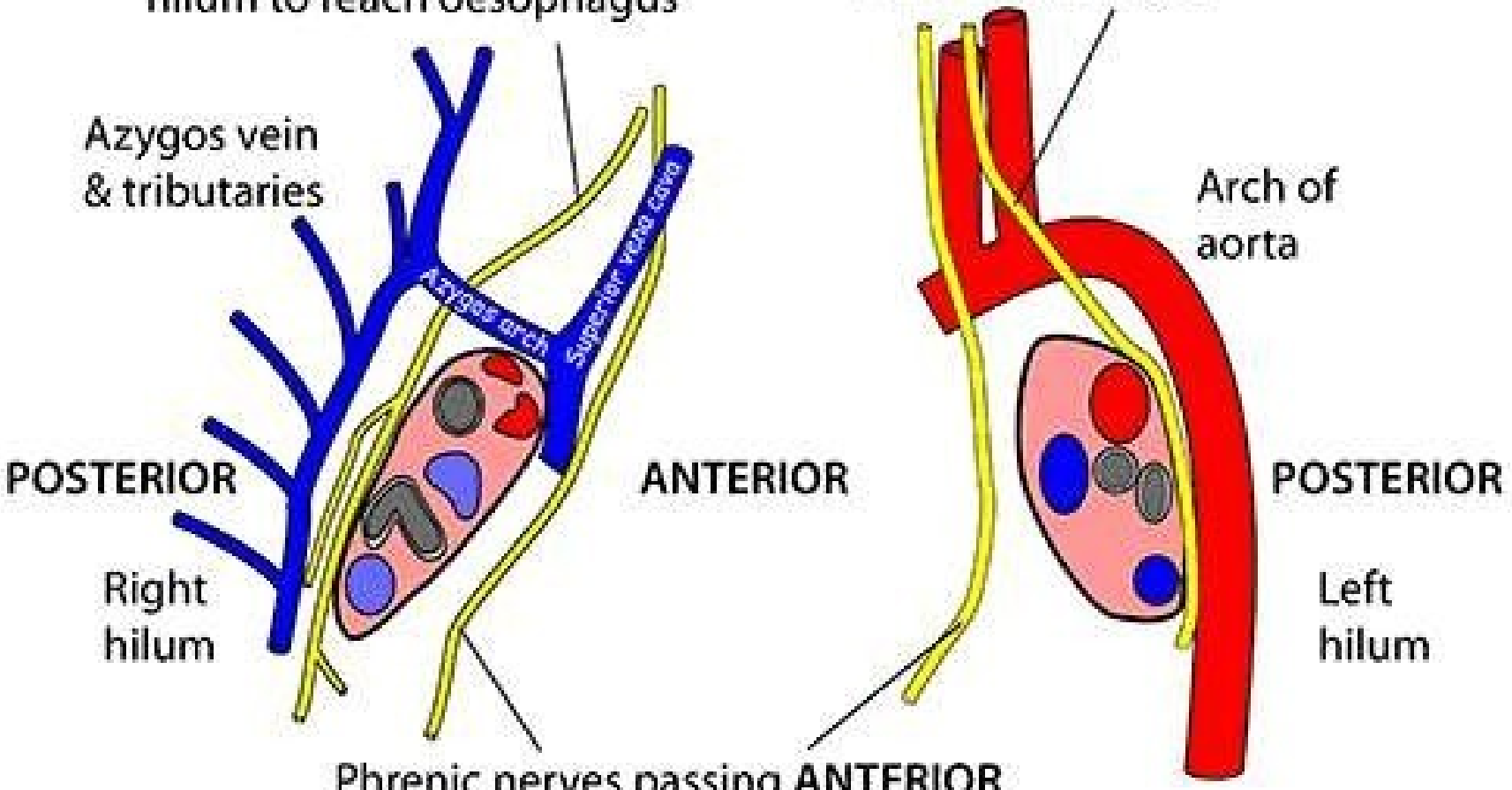
VAGUS NERVE

- **In the Thorax**
- In the thorax, the right vagus nerve forms the **posterior vagal trunk**, and the left forms the **anterior vagal trunk**.
- The vagal trunks enter the abdomen via the oesophageal hiatus.
  - Branches from the vagal trunks contribute to the formation of the oesophageal plexus, which innervates the smooth muscle of the oesophagus.
- Two other branches arise in the thorax:
  - **Left recurrent laryngeal nerve** – it hooks under the arch of the aorta, ascending to innervate the majority of the intrinsic muscles of the larynx.
  - **Cardiac branches** – these innervate regulate heart rate and provide visceral sensation to the organ.

RELATIONS OF PHRENIC & VAGUS NERVES TO HILA

Right vagus medial to azygos arch then **POSTERIOR** to hilum to reach oesophagus

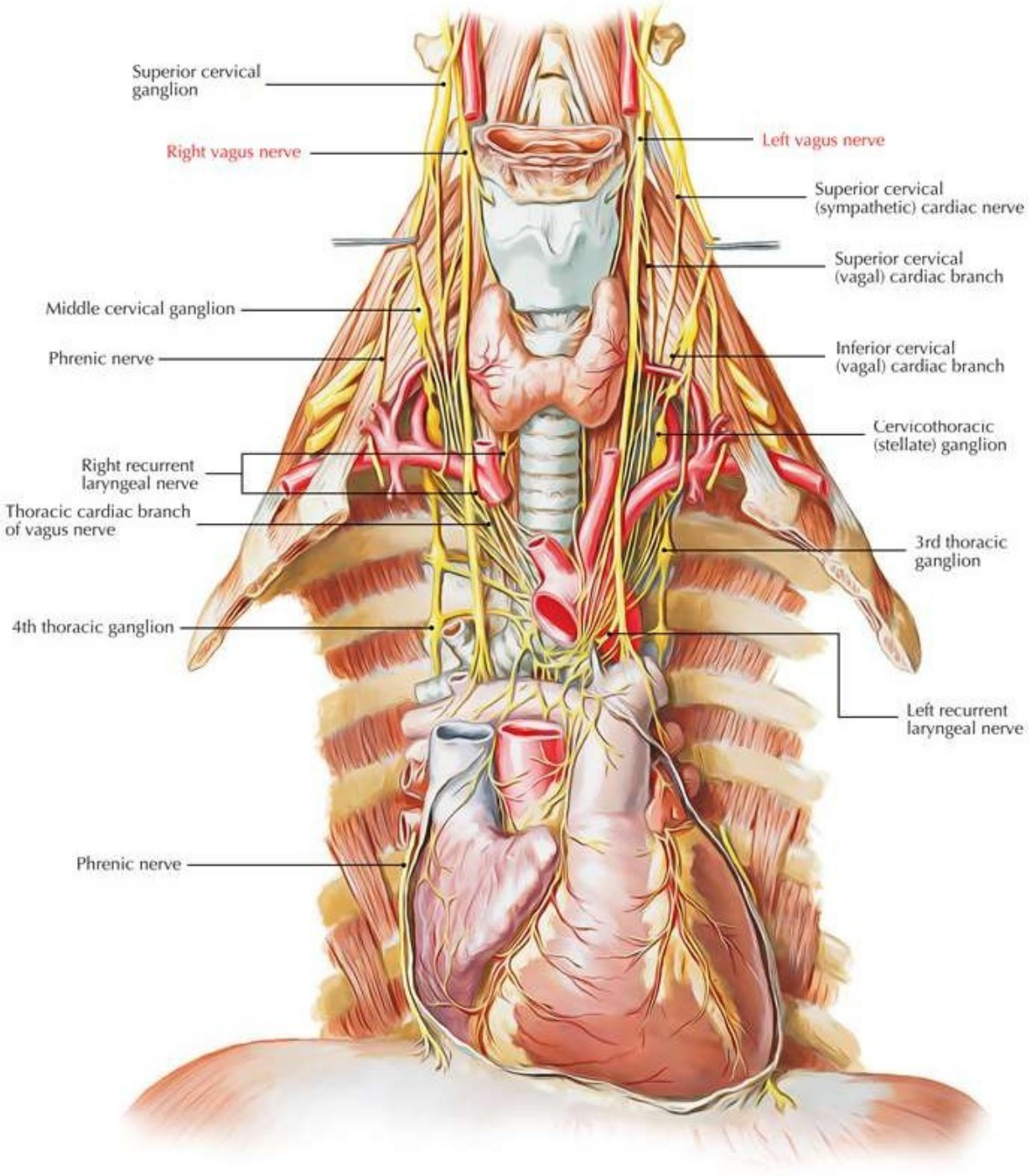
Left vagus over arch of aorta then **POSTERIOR** to hilum to reach oesophagus



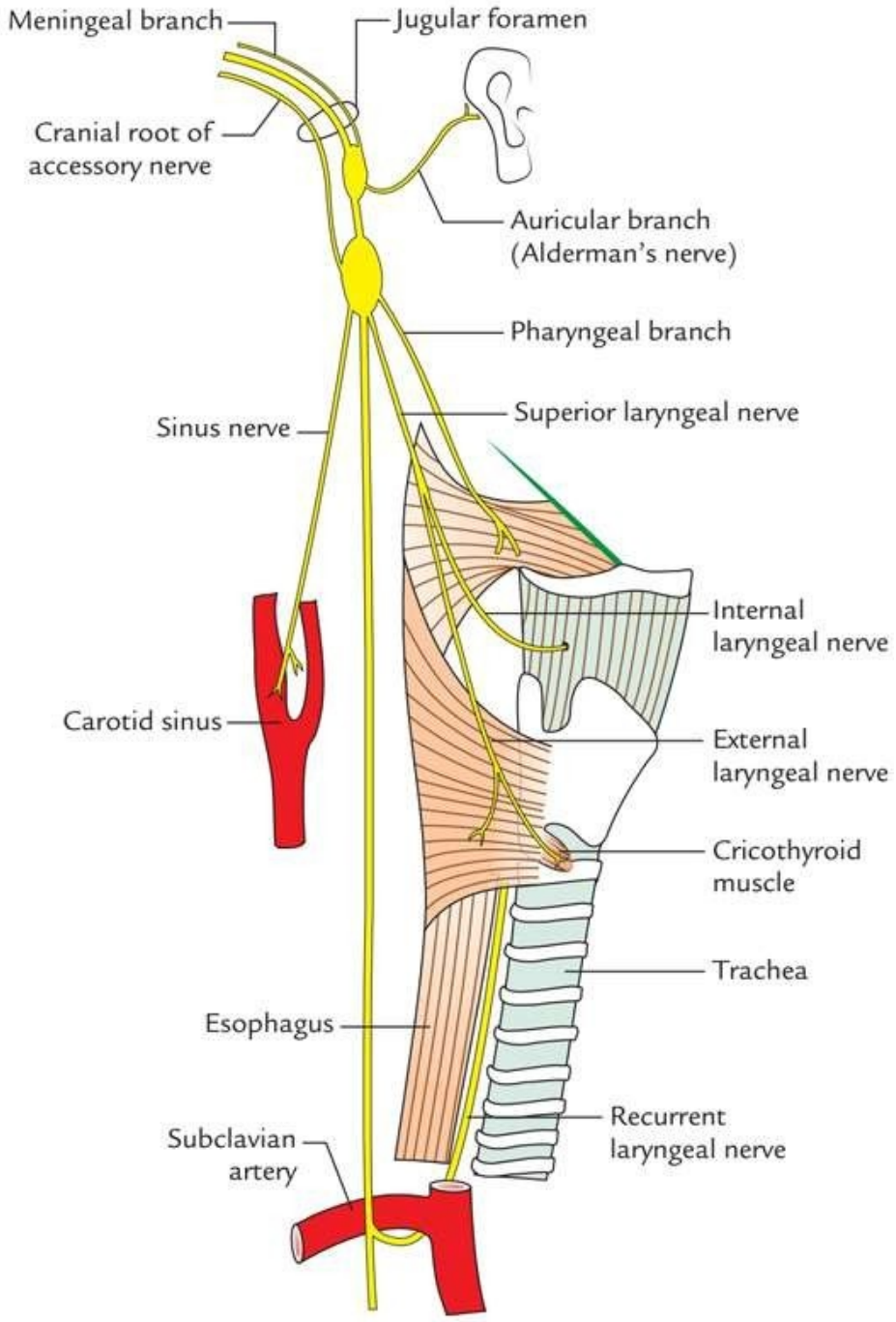
Vagus nerve

- **In the Abdomen**
- In the abdomen, the vagal trunks terminate by dividing into branches that supply:
- The oesophagus,
- The Stomach and
- The small and large bowel (up to the splenic flexure)

VAGUS NERVE


Superior cervical ganglion
Right vagus nerve
Left vagus nerve
Superior cervical (sympathetic) cardiac nerve
Superior cervical (vagal) cardiac branch
Middle cervical ganglion
Phrenic nerve
Inferior cervical (vagal) cardiac branch
Cervicothoracic (stellate) ganglion
Right recurrent laryngeal nerve
Thoracic cardiac branch of vagus nerve
3rd thoracic ganglion
4th thoracic ganglion
Left recurrent laryngeal nerve
Phrenic nerve

Vagus Nerve: Course and Connections
Vagus Nerve: Branches



VAGUS NERVE

- It supplies:
- All the muscles of pharynx with the exception of the stylopharyngeus that is supplied by the glossopharyngeal nerve and
- All the muscles of **soft palate** with the exception of the tensor palati that is supplied by the **mandibular nerve** (via the nerve to medial pterygoid).

Recurrent laryngeal nerve
(nerve of 6th arch)

- On the right side, it originates in the root of the neck from the vagus nerve.
- On the left side, it appears in the superior mediastinum from the vagus nerve as it crosses the arch of the **aorta**(lateral aspect). It hooks below the arch of the aorta on the left side of ligamentum arteriosum behind the arch of aorta on its way to the tracheoesophageal groove.
  - The recurrent laryngeal nerve gives motor innervation to all the intrinsic muscles of the larynx (with the exception of the cricothyroid that is supplied by the external laryngeal nerve) and sensory innervation to the mucous membrane of laryngeal cavity up to the vocal cord.
- Every recurrent laryngeal nerve enters deep to the inferior constrictor muscle to go into the laryngeal cavity deep to the cricothyroid joint. Now it's named the inferior laryngeal nerve.

LESIONS OF VAGUS NERVE

- The bilateral lesions of vagus nerve cause:
- nasal regurgitation of the swallowed liquids,
- nasal twang of voice,
- hoarseness of voice,
- flattering of palatal arches,
- cadaveric position of sung cards,
- dysphagia and
- reduction of cough reflex.

CLINICAL TESTING OF VAGUS NERVE

- The vagus nerve can be analyzed medically by requesting the patient to open his mouth and say ‘ah’ and after that comparing the palatal arches of the 2 sides. If the vagus is undamaged, the soft palate rises (is elevated) in the midline. In bilateral lesions, the soft palate falls. In the unilateral lesion, there’s flattening (drooping) of palate arch on the side of paralysis and uvula pulled to the normal side.